

QUESTION 1. Start on a new page

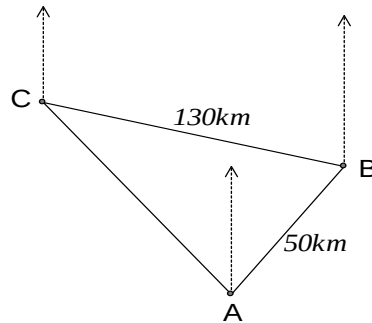
- (a) Find the value of $e^{2\pi}$ correct to 2 significant figures. 2
- (b) Factorise $3x^2 + x - 2$. 2
- (c) Find the value(s) of x for which $|2x - 1| = 5$. 2
- (d) Simplify $\frac{1}{2} - \frac{1}{x+1}$. 2
- (e) If $\frac{20\sqrt{18}}{16\sqrt{24}} = a\sqrt{3}$ find the value of a . 2
- (f) If α and β are the roots of the equation $x^2 + 6x - 3 = 0$
evaluate $\alpha^2\beta + \beta^2\alpha$. 2

QUESTION 2. Start on a new page.

- (a) Differentiate :
- (i) $(e^x + 2)^8$ 2
- (ii) $\frac{x-1}{\cos x}$ 2
- (b) Find:
- (i) $\int_0^{\frac{\pi}{3}} \sin 4x \, dx$ 3
- (ii) $\int \frac{1-x^2}{x} \, dx$ 3
- (c) Solve $2\log_3 5 - \log_3 x = 2$ 2

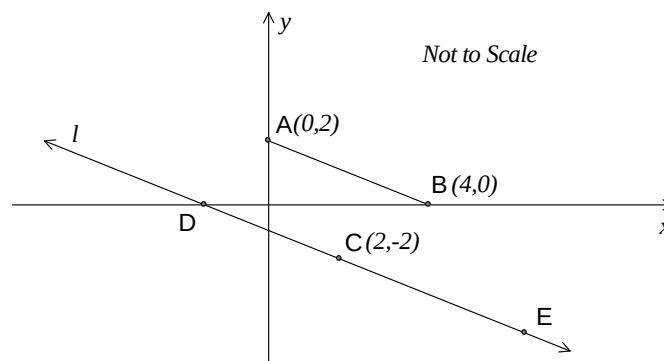
QUESTION 3. Start on a new page.

- (a) A ship sails 50 km from port A to port B on a bearing of 063° . It then sails 130 km from port B to port C on a bearing of 296° .



- (i) Show (with working) $\angle ABC = 53^\circ$ 1
- (ii) Find AC to the nearest kilometre. 2

(b)



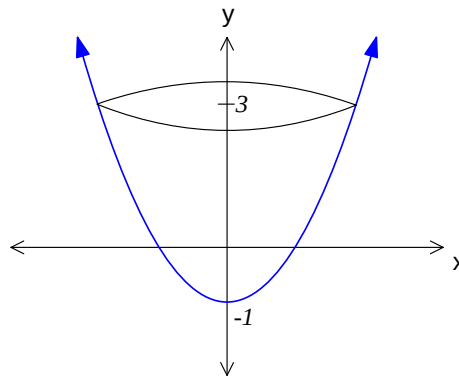
- (i) Using the diagram above find the gradient of AB. 1
- (ii) Show that the distance AB is $2\sqrt{5}$ units. 1
- (iii) Show that the equation of the line l , through C parallel to AB is $x + 2y + 2 = 0$. 2
- (iv) Find the co-ordinates of the point D where l cuts the x axis. 1
- (v) Show that the perpendicular distance from A to the line l is $\frac{6}{\sqrt{5}}$ units. 1
- (vi) If the area of the trapezium is 18 $units^2$ find the co-ordinates of the point E. 3

QUESTION 4. Start on a new page.

- (a) For what value(s) of k does the equation $x^2 - (k + 2)x + 2k + 1 = 0$ have real roots. 3
- (b) Let $f(x) = x(x - 4)^3$.
- (i) Show that $f'(x) = 4(x - 4)^2(x - 1)$. 2
- (ii) Find the co-ordinates of the stationary points and determine their nature. 3
- (iii) If $f''(x) = 12(x - 2)(x - 4)$ find the points of inflexion on the curve. 2
- (iv) Sketch the curve $y = f(x)$ showing intercepts, stationary points and points of inflexion. 2

QUESTION 5. Start on a new page.

- (a) (i) Find the probability that in a family of 4 children 2 are boys and 2 are girls. (You may wish to draw a tree diagram). 2
- (ii) Two women each have had 4 children. What is the probability that neither woman has 2 boys and 2 girls? 2
- (b) The shape of a glass is formed by rotating the curve $y = 2x^2 - 1$ about the y axis from $y = -1$ to $y = 3$.



Find the volume of the glass. 3

- (c) In 1990 a farmer harvested 900 tonnes of wheat. This amount increased by 4% each year.
- (i) How much was harvested in the year 2000? 2
- (ii) How much was harvested from 1990 to 2000? 3

QUESTION 6. Start on a new page.

(a) (i) Show that $\sin \theta \cot \theta = \cos \theta$ 1

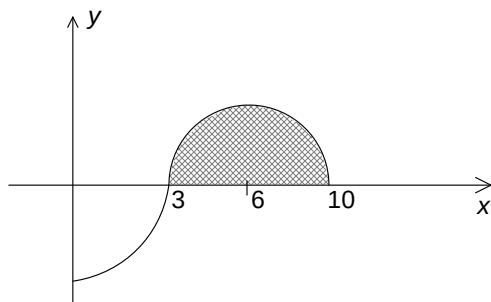
(ii) Hence solve $27 \cos \theta \sin \theta \cot \theta = \sec \theta$ 3

(b) (i) If $y = \sqrt{\sin x}$ complete the table. 1

x	0	0.5	1	1.5	2
y					0.95

(ii) Hence evaluate $\int_0^2 \sqrt{\sin x} \, dx$ using Simpson's rule with 5 function values. 3

(c) The graph shows the velocity- time graph of an object as it moves along the x axis in 10 seconds.



(i) When does the object come to rest? 1

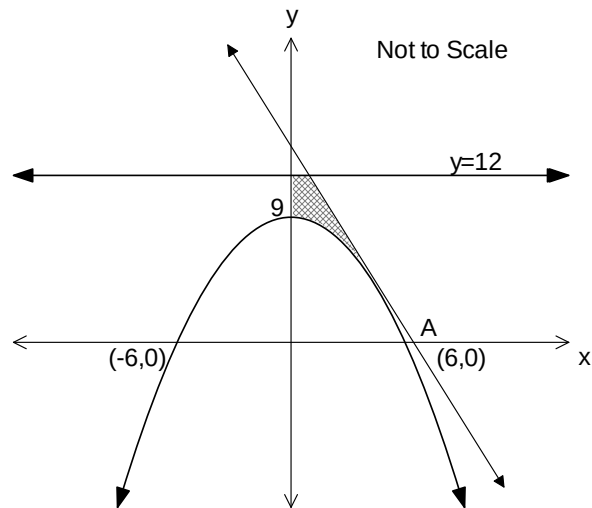
(ii) When is the acceleration of the object the greatest? 1

(iii) What does the shaded region represent? 1

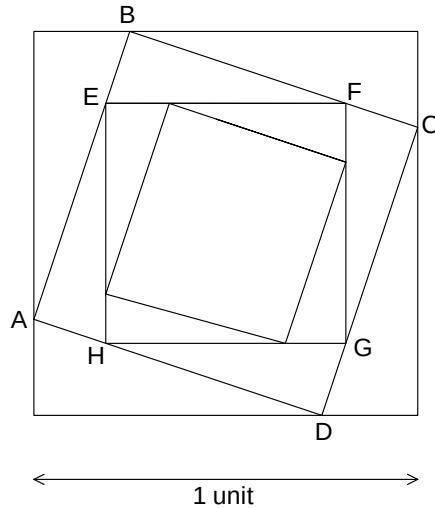
(iv) At approximately what time will the object return to its starting point? 1

QUESTION 7. Start on a new page.

(a) The graph of $y = 9 - \frac{x^2}{4}$ is drawn below. The tangent to the curve at A(6,0) is also drawn

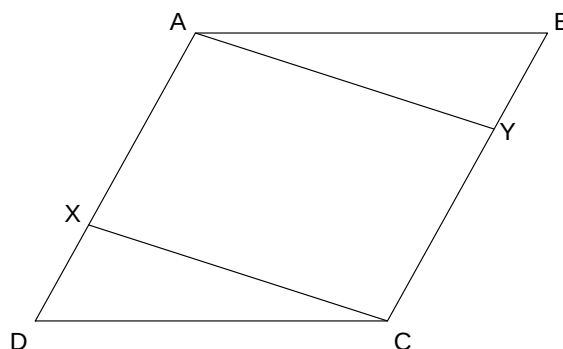


- (i) Show that the equation of the tangent to the curve at A is given by $y = -3x + 18$. 2
- (ii) What are the co-ordinates of the focus of the parabola $y = 9 - \frac{x^2}{4}$? 2
- (iii) Find the area of the shaded region. 4



- (b) In the diagram above the largest square has sides 1 unit. This pattern of squares has consecutive squares whose dimensions are $\frac{3}{4}$ that of the previous square. For example square ABCD has side lengths of $\frac{3}{4}$ of a unit.

- (i) What is the area of square EFGH? 1
- (ii) What is the sum of the area of all the squares if this pattern continued

QUESTION 8. Start on a new page.

- (a) ABCD is a rhombus. X and Y lie on AD and BC such that $XC \perp AD$ and $AY \perp BC$.

(i) Prove $\triangle ABC \equiv \triangle DCX$. 2

(ii) If the side length of the rhombus is 10 cm and $\angle DAB = 120^\circ$, find the length of the diagonal AC. 2

(iii) Find the area of the rectangle AYCX. 2

- (b) The rate (R) at which green house gases are released into the atmosphere from

a town in tonnes/hour is given by: $R = 20 + \frac{100}{(1+t)^2}$, where t is in hours.

(i) At what rate are the green house gases released initially? 1

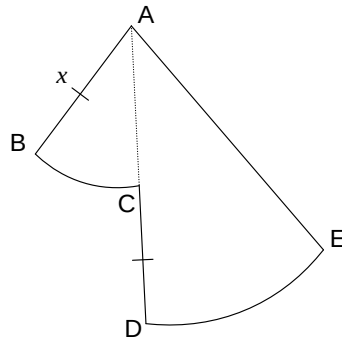
(ii) What is the rate at which green house gases are released as time increases indefinitely? 1

(iii) Without using calculus draw a graph of R as a function of time. 1

(iv) How much gas was released into the atmosphere in the first 2 hours? 3

QUESTION 9. Start on a new page.

- (a) Two sectors make up a company logo.



Both sectors have centre A , $AB=CD$, $AB=x$ and AC bisects $\angle BAE$.

Let $\angle BAC = \theta$.

(i) If the area of the logo is $8\pi \text{ units}^2$ show that: $\theta = \frac{16\pi}{5x^2}$ 2

(ii) Show that the perimeter (P) of the logo is given by

$$P = 4x + \frac{48\pi}{5x} \quad 2$$

(iii) Find the value of x which makes the perimeter of the logo a minimum. 3

(b) (i) Sketch the curve $y = 1 - 3\cos 2x$ for $-\pi \leq x \leq \pi$. 3

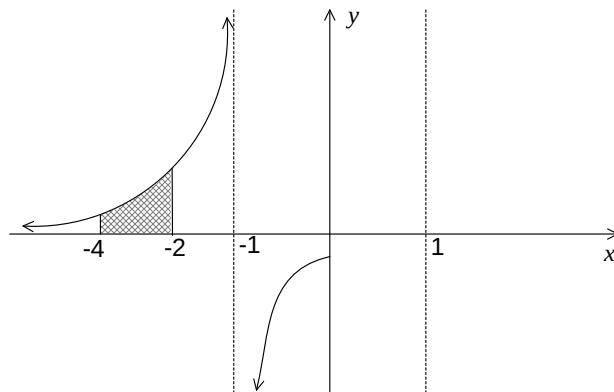
(ii) If $3\cos 2x = 1 - k$ has 3 solutions in the given domain, find the value of k . 2

QUESTION 10. Start on a new page.

(a) (i) Show that the derivative of $\log_e \frac{x-1}{x+1}$ is $\frac{2}{(x-1)(x+1)}$ 2

(ii) Show that $f(x) = \frac{1}{(x-1)(x+1)}$ is an even function. 2

(iii)



A portion of the graph of $f(x) = \frac{1}{(x-1)(x+1)}$ is drawn above.

Find the shaded area.

3

(b) Evaluate $\sum_{k=4}^{20} 2(2^{k-4} + 2k - 3)$

5

3 b) (i) $m_{AB} = -\frac{1}{2}$ ①
 (ii) distance $AB = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$ ①

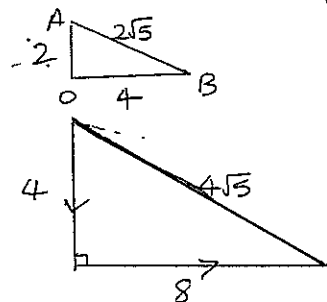
(iii) $y+2 = -\frac{1}{2}(x-2)$ ①
 $2y+4 = -x+2$ no use.
 $x+2y+2=0$

(iv) at D $y=0$
 $\therefore x+0+2=0$
 $\therefore x=-2$ ①
 $D(-2, 0)$

(v) $d = \frac{|0(1)+2(2)+2|}{\sqrt{1^2+2^2}}$ ①
 $= \frac{6}{\sqrt{5}}$

(vi) $\frac{1}{2} \times \frac{6}{\sqrt{5}} (a+2\sqrt{5}) = 18$
 $a = 4\sqrt{5}$ ①
 $a = DE$
 $\triangle ABO \parallel \triangle DFE$

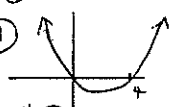
P AB: DE = 1:2 ①



$\therefore E(-2+8, 0-4)$
 $E(6, -4)$ ①

Question 4

a) Real roots when $\Delta \geq 0$
 $[-(k+2)]^2 - 4(2k+1) \geq 0$ ①
 $k^2 + 4k + 4 - 8k - 4 \geq 0$
 $k^2 - 4k \geq 0$
 $k(k-4) \geq 0$ ①
 $\therefore k \leq 0, k \geq 4$ ①



b) (i) $f(x) = x(x-4)^3$
 $f'(x) = 1(x-4)^3 + 3(x-4)^2 \cdot x$ ①
 $= (x-4)^2 [(x-4) + 3x]$
 $= 4(x-4)^2(x-1)$ ①

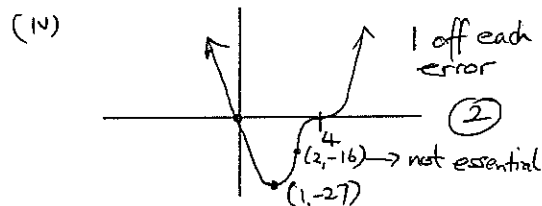
(ii) st. pts when $y=0$
 $4(x-4)^2(x-1) = 0$ when
 $x=1$ $x=4$
 $y=-27$ $y=0$
 $(1, -27)$ $(4, 0)$ ①

Test:

	0	1	2	4	5
y'	$4 \times 16 = 64$	0	$4 \times 16 = 64$	0	$4 \times 16 = 64$
	↗	↘	↗	↘	↗

Min ①
 Horiz. inflex. ①

(ii) Pt of inflexion at $(4, 0)$ (part i)
 $y''=0$ $12(x-2)(x-4)$
 $x=2$ $y=-16$ $(2, -16)$ ①
 Test concavity change.
 When $x=1$ $y''=+36$ ①
 When $x=3$ $y''=-12$
 Concavity change $\therefore$ p.s.o.i.
 $(2, -16)$ & $(4, 0)$



1. a) $e^{2\pi} = 535.49...$ ①
 $= 540$

If error must show full working to get ①

b) $(3x-2)(x+1)$ ②

① mark for $(3x+2)(x-1)$
 ± 1 ± 2

c) $|2x-1|=5$
 $2x-1=5$ $-2x+1=5$
 $x=3$ ① $x=-2$ ①

d) $\$979 = 110\%$ of org.
 $1\% = \frac{979}{110}$ ①
 Org. Price = $\frac{979}{110} \times 100$
 $= \$890$ ①

e) $\frac{20\sqrt{18}}{16\sqrt{24}} = \frac{5\sqrt{3}}{4\sqrt{4}}$
 $= \frac{5\sqrt{3}}{8}$

$a = \frac{5}{8}$ ②
 1 off for not simplifying.

Solutions:- 2U 09 Trial

f) $\alpha + \beta = -6$ $\alpha\beta = -3$

①
 $\alpha^2\beta + \beta^2\alpha = \alpha\beta(\alpha + \beta)$
 $= -3(-6)$
 $= 18$ ①

2a) (i) $y = (e^x + 2)^8$
 $y' = 8e^x(e^x + 2)^7$
 ① ②

(ii) $y = \frac{x-1}{\cos x}$ 1 off each error
 $y' = \frac{1 \cdot \cos x + \sin x(x-1)}{\cos^2 x}$ ②
 $= \frac{\cos x + x \sin x - \sin x}{\cos^2 x}$ (no use if they cancel)

b) $\int_0^{\frac{\pi}{3}} \sin 4x \, dx = -\frac{1}{4} [\cos 4x]_0^{\frac{\pi}{3}}$ ①
 $= -\frac{1}{4} [\cos \frac{4\pi}{3} - \cos 0]$ ①
 $= -\frac{1}{4} [-\frac{1}{2} - 1]$
 $= \frac{3}{8}$ ①

(i) $\int \frac{1-x^2}{x} \, dx = \int \frac{1}{x} - x \, dx$ ①
 $= \ln x - \frac{x^2}{2} + C$
 ① ① (ignore C)

c) $2 \log_3 5 - \log_3 2 = 2$
 $\log_3 25 - \log_3 2 = 2$
 $\log_3 \left(\frac{25}{2}\right) = 2$ ①
 $3^2 = \frac{25}{2}$
 $x = \frac{25}{9}$ ①

3a) $\angle NAB = 63^\circ$
 $\angle CBX = 64^\circ$ ($\angle$'s at a pt)
 $\angle ABC = 53^\circ$ (Co-interior $\angle$'s on 11 lines)
 ① accept 180 - (63+64) not essential to have reasons.
 (ii) $AC^2 = 50^2 + 130^2 - 2 \cdot 50 \cdot 130 \cdot \cos 53$ ①
 $= 107.59$
 $= 108 \text{ km.}$ ①

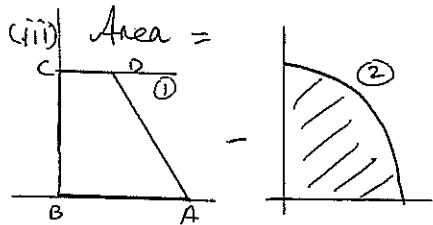
$$7a) (i) \quad y = 9 - \frac{x^2}{4}$$

$$\frac{x^2}{4} = -y + 9$$

$$x^2 = -4(y - 9) \quad (1)$$

$$\therefore 4a = 4 \quad a = 1$$

$$\text{Focus } (0, 8) \quad (1)$$



$$\text{at } D \quad y = 12 \text{ \& } y = -3x + 18$$

$$\therefore 12 = -3x + 18$$

$$x = 2$$

$$\therefore \text{Area of trap.} = \frac{1}{2} \cdot 12(2+6)$$

$$= 48 \text{ units}^2 \quad (1)$$

Area ② $\Rightarrow$ make x the subject

$$x^2 = -4y + 36$$

$$x = \sqrt{36 - 4y}$$

$$= 2(9 - y)^{\frac{1}{2}} \quad (1)$$

$$\therefore \text{Area} = 2 \int_0^9 (9 - y)^{\frac{1}{2}} dy$$

$$= 2 \left[-\frac{2}{3} (9 - y)^{\frac{3}{2}} \right]_0^9 \quad (1)$$

$$= -\frac{4}{3} \left[(9 - y)^{\frac{3}{2}} \right]_0^9$$

$$= -\frac{4}{3} [0 - 27]$$

$$= \frac{108}{3} = 36$$

$$\therefore \text{Shaded area} = 48 - 36 \quad (1)$$

(many ways of doing this) $= 12 \text{ units}^2$

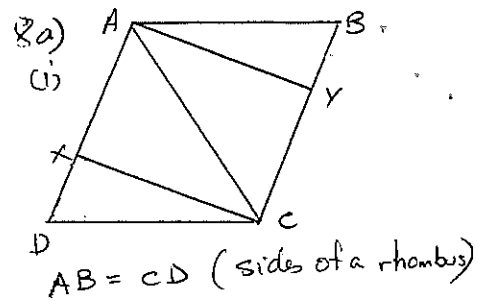
$$b) (i) \left(\frac{3}{4} \times \frac{3}{4} \right)^2 = \frac{81}{256} \text{ units}^2 \quad (1)$$

$$(ii) \text{Area of squares} =$$

$$1 + \frac{9}{16} + \frac{81}{256} + \dots \quad (1)$$

$$S_{\infty} = \frac{1}{1 - \frac{9}{16}} \quad (1)$$

$$= \frac{16}{7} \text{ units}^2 \quad (1)$$



$$\angle AYB = 90^\circ \quad (AY \perp BC) \quad (1)$$

$$\angle AXC = 90^\circ \quad (AD \perp XC) \quad (1)$$

$$\angle B = \angle D \quad (\text{opp. } \angle \text{s of a rhombus})$$

$$\therefore \triangle ABY \equiv \triangle DXC \quad (\text{AAS}) \quad (1)$$

(ii) Construct AC

$$\angle DAC = 60^\circ \quad (\text{diagonal bisects } \angle \text{ at the corner})$$

$$\angle ADC = 60^\circ \quad (\text{constr. } \angle \text{s on 11 line})$$

$$\therefore \triangle ADC \text{ is equilateral} \quad (1)$$

$$\therefore AC = 10 \text{ cm.} \quad (\text{not essential})$$

$$(iii) AX = 5 \text{ cm} \quad (\text{Perpendicular bisects base})$$

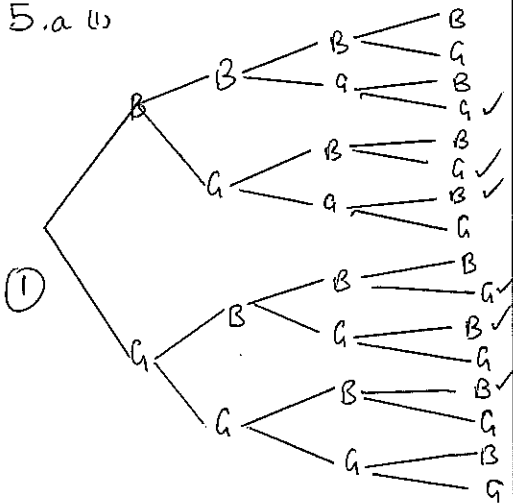
$$\sin 60^\circ = \frac{XC}{10}$$

$$\therefore XC = 10 \cdot \frac{\sqrt{3}}{2} \quad (1)$$

$$= 5\sqrt{3}$$

$$\therefore \text{Area} = 5 \times 5\sqrt{3} = 25\sqrt{3}$$

5.a) (i)



$$P(2B2G) = \frac{6}{16} = \frac{3}{8} \quad (1)$$

$$(ii) \left(1 - \frac{3}{8} \right) \left(1 - \frac{3}{8} \right) = \frac{55}{64} \quad (1)$$

$$b) \quad y = 2x^2 - 1$$

$$x^2 = \frac{y+1}{2}$$

$$V = \pi \int_{-1}^3 \frac{y}{2} + \frac{1}{2} dy \quad (1)$$

$$= \frac{\pi}{2} \left[\frac{y^2}{2} + y \right]_{-1}^3 \quad (1)$$

$$= \frac{\pi}{2} \left[\left(\frac{9}{2} + 3 \right) - \left(\frac{1}{2} - 1 \right) \right]$$

$$= 4\pi \text{ units}^3 \quad (1)$$

$$c) (i) 900 \times 1.04^{10} \rightarrow (1)$$

$$= 1332.22 \text{ \&}. \quad (1)$$

$$(ii) \text{Total} = 900 + 900 \times 1.04 + \dots + 900 \times 1.04^{10} \quad (1)$$

$$\text{G.P. } a = 900 \quad r = 1.04 \quad n = 11$$

$$S_n = \frac{900(1.04^{11} - 1)}{1.04 - 1} \quad (1)$$

$$= 12137.72 \text{ \&}. \quad (1)$$

$$(a) (i) \text{ L.H.S. } \sin \theta \cot \theta$$

$$= \sin \theta \cdot \frac{\cos \theta}{\sin \theta} \quad (1)$$

$$= \cos \theta$$

$$(ii) 27 \cos \theta \cot \theta \sin \theta = \sec \theta$$

$$\textcircled{1} 27 \cos \theta \cdot \cos \theta = \frac{1}{\cos \theta}$$

$$27 \cos^3 \theta = 1$$

$$\cos^3 \theta = \frac{1}{27}$$

$$\cos \theta = \frac{1}{3} \quad (1)$$

$$\theta = 70^\circ 32' \quad (1)$$

$$\text{or } \theta = 1.23 \text{ radians}$$

$$y = \sqrt{\sin x}$$

x	0	0.5	1	1.5	2
y	0	0.69	0.92	1.0	0.95

$$A = \frac{0.5}{3} \left[0 + 0.95 + 4(0.69 + 1) + 2(0.92) \right] \quad (1)$$

$$= 1.5916^\circ \quad (1)$$

$$= 1.6$$

$$c) (i) t = 3, 10 \quad (1)$$

$$(ii) t = 3 \quad (1)$$

(iii) The distance travelled by the object between $t = 3$ and $t = 10$ (1)

(iv) $t = 6$, because total velocity to the left (1)
 $=$ " " to the right or
 $=$ distance travelled left + right equal at $t = 6$.

$$7a) \quad y = 9 - \frac{x^2}{4}$$

$$y' = -\frac{2x}{4}$$

$$\text{at } x = 6 \quad y' = -3 \quad (1)$$

$$\therefore \text{Tangent } y - 0 = -3(x - 6) \quad (1)$$

$$y = -3x + 18$$

$$10a) (i) y = \ln\left(\frac{x-1}{x+1}\right)$$

$$y = \ln(x-1) - \ln(x+1) \quad (1)$$

$$\frac{dy}{dx} = \frac{1}{x-1} - \frac{1}{x+1} \quad (1)$$

$$= \frac{x+1 - (x-1)}{(x-1)(x+1)} \quad (1)$$

$$= \frac{2}{(x-1)(x+1)}$$

$$(ii) f(x) = \frac{1}{(x-1)(x+1)}$$

$$f(-x) = \frac{1}{(-x-1)(-x+1)} \quad (1)$$

$$= \frac{1}{-(x+1) - (x-1)} \quad (1)$$

$$= \frac{1}{(x+1)(x-1)}$$

$$f(x) = f(-x) \therefore \text{Even}$$

$$\int_{-4}^{-2} \frac{dx}{(x-1)(x+1)} = \int_2^4 \frac{dx}{(x-1)(x+1)} \quad (1)$$

since $y=f(x)$ is even

$$\int_2^4 \frac{dx}{(x-1)(x+1)} = \frac{1}{2} \left[\ln\left(\frac{x-1}{x+1}\right) \right]_2^4 \quad (1)$$

$$= \frac{1}{2} \left[\ln\left(\frac{3}{5}\right) - \ln\left(\frac{1}{3}\right) \right]$$

$$= \frac{1}{2} \ln\left(\frac{3 \times 3}{5 \times 1}\right) \quad (1)$$

$$= \frac{1}{2} \ln\left(\frac{9}{5}\right) = 0.2938 \dots$$

$$b) \sum_{k=4}^{20} 2(2^{k-4} + 2k-3)$$

$$= 2 \left[\sum_{k=4}^{20} 2^{k-4} + \sum_{k=4}^{20} 2k-3 \right] \quad (1)$$

$$= 2 \left[1+2+\dots+2^{16} \right] + 2 \left[5+7+\dots+37 \right] \quad (1)$$

$$2 \left[\frac{1(2^{17}-1)}{2-1} \right] + 2 \left[\frac{17(5+37)}{2} \right]$$

$$= 262142 \quad (1) + 714 \quad (1)$$

$$= 262856$$

for 10b)

(1) - for substitution

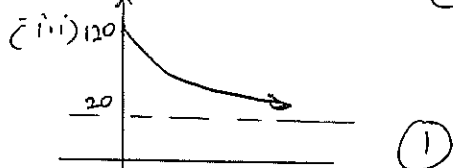
(1) - for $n=17$.

$$b) (i) R = 20 + \frac{100}{(1+2t)^2}$$

$$t=0 \quad R = 20 + \frac{100}{1} = 120 \text{ t/hr} \quad (1)$$

$$(ii) \text{ as } t \rightarrow \infty \quad \frac{100}{(1+2t)^2} \rightarrow 0$$

$$\therefore \text{as } t \rightarrow \infty \quad R = 20 \text{ t/hr} \quad (1)$$



$$(iv) Q_{\text{gas}} = \int_0^2 \left(20 + \frac{100}{(1+2t)^2} \right) dt \quad (1)$$

$$= \left[20t + \frac{100(1+2t)^{-1}}{-2} \right]_0^2$$

$$= \left[20t - \frac{50}{1+2t} \right]_0^2 \quad (1)$$

$$= \left(40 - \frac{50}{5} \right) - \left(0 - 50 \right)$$

$$= 80 \text{ tonnes.} \quad (1)$$

$$9 (i) \text{ Area} = \frac{1}{2} x^2 \theta + \frac{1}{2} (2x)^2 \theta \quad (1)$$

$$= \frac{1}{2} x^2 \theta + 2x^2 \theta$$

$$A = \frac{5x^2 \theta}{2}$$

$$\text{If } A = 8\pi \quad 8\pi = \frac{5x^2 \theta}{2} \quad (1)$$

$$\theta = \frac{16\pi}{5x^2} \quad (1)$$

$$(ii) P = x + x\theta + x + 2x\theta + 2x \quad (1)$$

$$= 4x + 3x\theta$$

$$P = 4x + 3x \cdot \frac{16\pi}{5x^2} \quad (1)$$

$$= 4x + \frac{48\pi}{5x}$$

$$(iii) P' = 4 - \frac{48\pi}{5} x^{-2} \quad (1)$$

$$= 4 - \frac{48\pi}{5x^2} \quad (1)$$

Min. when $P' = 0 \quad P'' > 0$

$$4 - \frac{48\pi}{5x^2} = 0$$

$$20x^2 = 48\pi$$

$$x^2 = \frac{48\pi}{20} = \frac{12\pi}{5}$$

$$\therefore x = \pm \sqrt{\frac{12\pi}{5}} \quad \text{but } x > 0$$

$$\therefore x = \sqrt{\frac{12\pi}{5}} \quad (1)$$

$$\text{Test } P'' = \frac{96\pi x}{5}$$

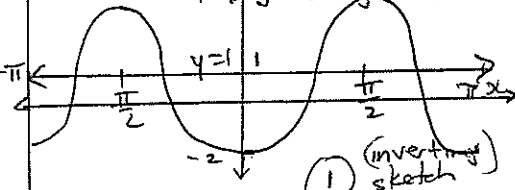
$$\text{when } x = \sqrt{\frac{12\pi}{5}} \quad P'' = \frac{96\pi}{5} \cdot \sqrt{\frac{12\pi}{5}} > 0 \quad (1)$$

$\therefore$ Min. occurs

$$\text{when } x = \sqrt{\frac{12\pi}{5}} =$$

$$(iv) (i) y = 1 - 3\cos 2x$$

$$\text{Amp} = 3 \quad (1) \quad \text{Period} = \pi \quad (1)$$



$$(ii) 1 - 3\cos 2x = k \quad (1)$$

ie $3\cos 2x = 1 - k$
will have 3 solutions if $k = -2$ (answer only) $(1) \quad (1/2)$

Because $y = -2$ & $y = 1 - 3\cos 2x$ have 3 pts of intersection.